

PHASE 7.2

STATIC EXPLANATION BASELINE AND EVALUATION TASKS

Comparison Material for the Static Baseline and the Interactive Power BI Dashboard

Master's Thesis Title:

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

1. Purpose of this Document

This document provides the fixed visual baseline and the task set used to compare a non-interactive presentation with the completed Power BI dashboard. The static condition contains screenshots of the six final dashboard pages. Participants can read the information shown, but they cannot change filters, select another customer, or interact with the visual elements.

The interactive condition uses the same underlying data and model outputs through the live Power BI report. The questions are kept equivalent across both conditions so that task accuracy and completion time can be compared.

2. Participant Instructions

- Complete the task set for the assigned condition without receiving hints.
- Record the start time before Task 1 and the finish time after Task 6.
- For the static condition, use only the screenshots included in this document.
- For the interactive condition, use the live Power BI report and reset all filters before starting.
- Record one answer for Part A and one answer for Part B in every task.
- Do not view the answer key until both conditions have been completed.

3. Task Coverage

The six tasks cover the complete analytical flow of the dashboard:

- Executive summary and prediction totals
- Customer risk identification
- Global SHAP interpretation
- Individual customer explanation
- Customer segment comparison
- Model performance and threshold interpretation

4. Static View 1 - Executive Overview

This view summarises the number of customers, actual and predicted churn, the selected threshold, overall model metrics, churn distributions, and the confusion matrix.

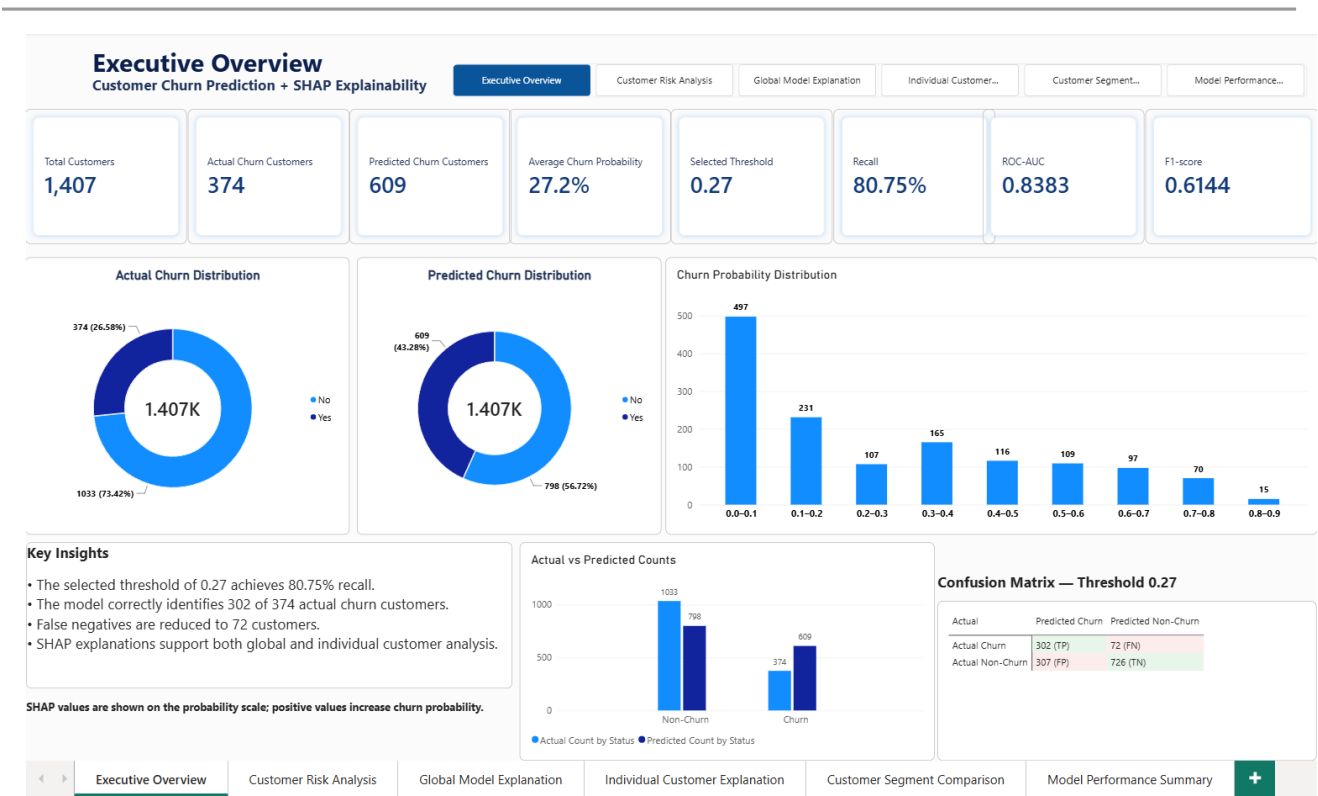


Figure 1. Fixed screenshot used in the static explanation condition.

Relevant task: Task 1 - Executive Overview

5. Static View 2 - Customer Risk Analysis

This view presents customer risk categories, the highest-risk customers, tenure and churn probability, and the detailed customer risk table.

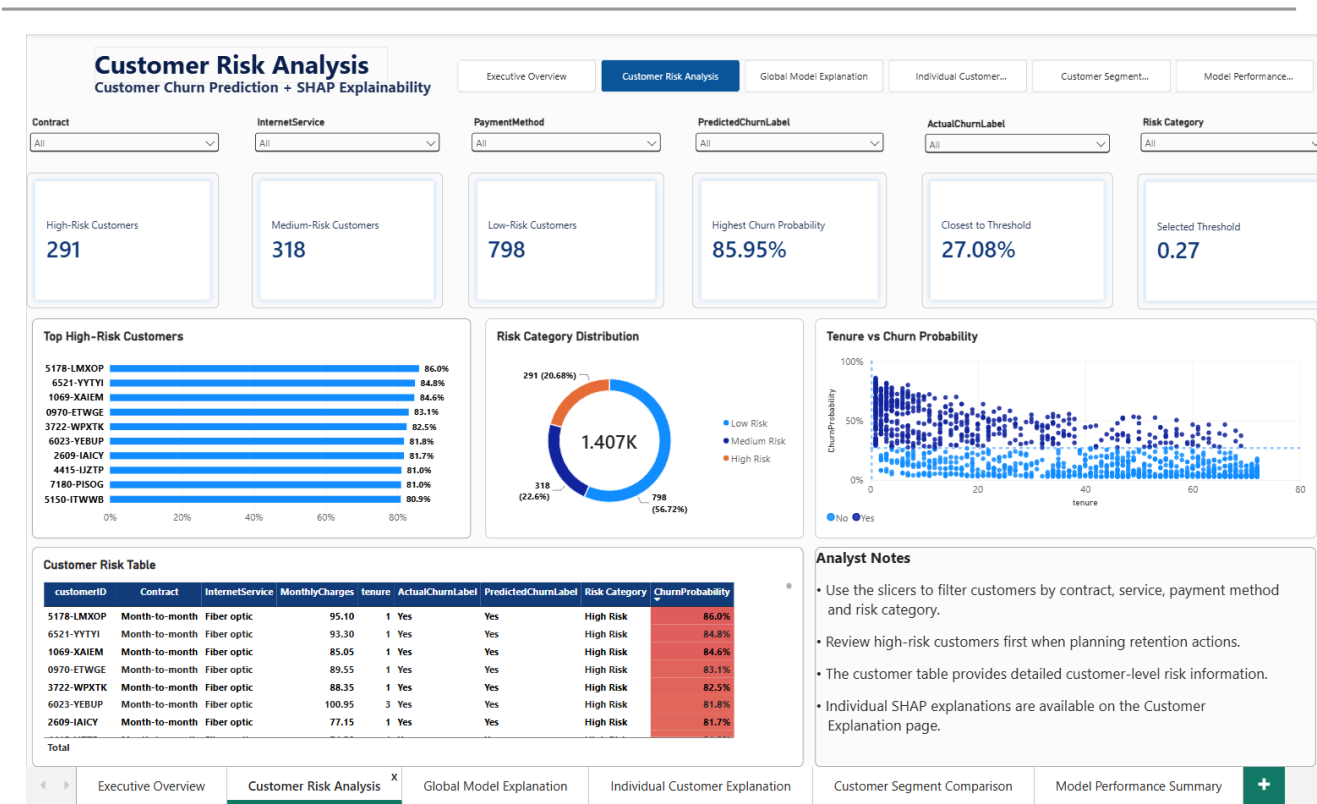


Figure 2. Fixed screenshot used in the static explanation condition.

Relevant task: Task 2 - Customer Risk Analysis

6. Static View 3 - Global Model Explanation

This view presents grouped global SHAP importance, average direction of influence, the top global features, and numerical feature relationships.

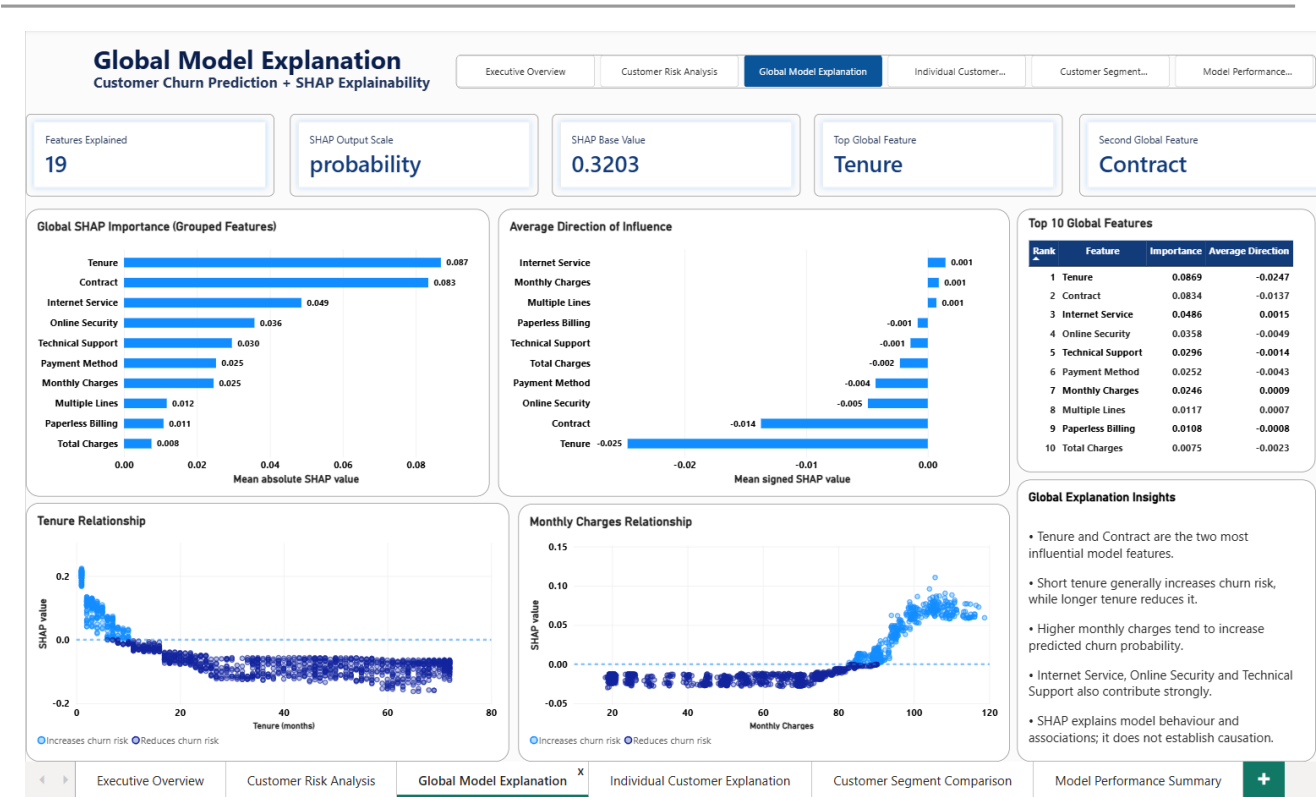


Figure 3. Fixed screenshot used in the static explanation condition.

Relevant task: Task 3 - Global Model Explanation

7. Static View 4 - Individual Customer Explanation

This view explains the prediction for customer 5178-LMXOP using the customer profile, churn probability, risk gauge, and local SHAP contributions.

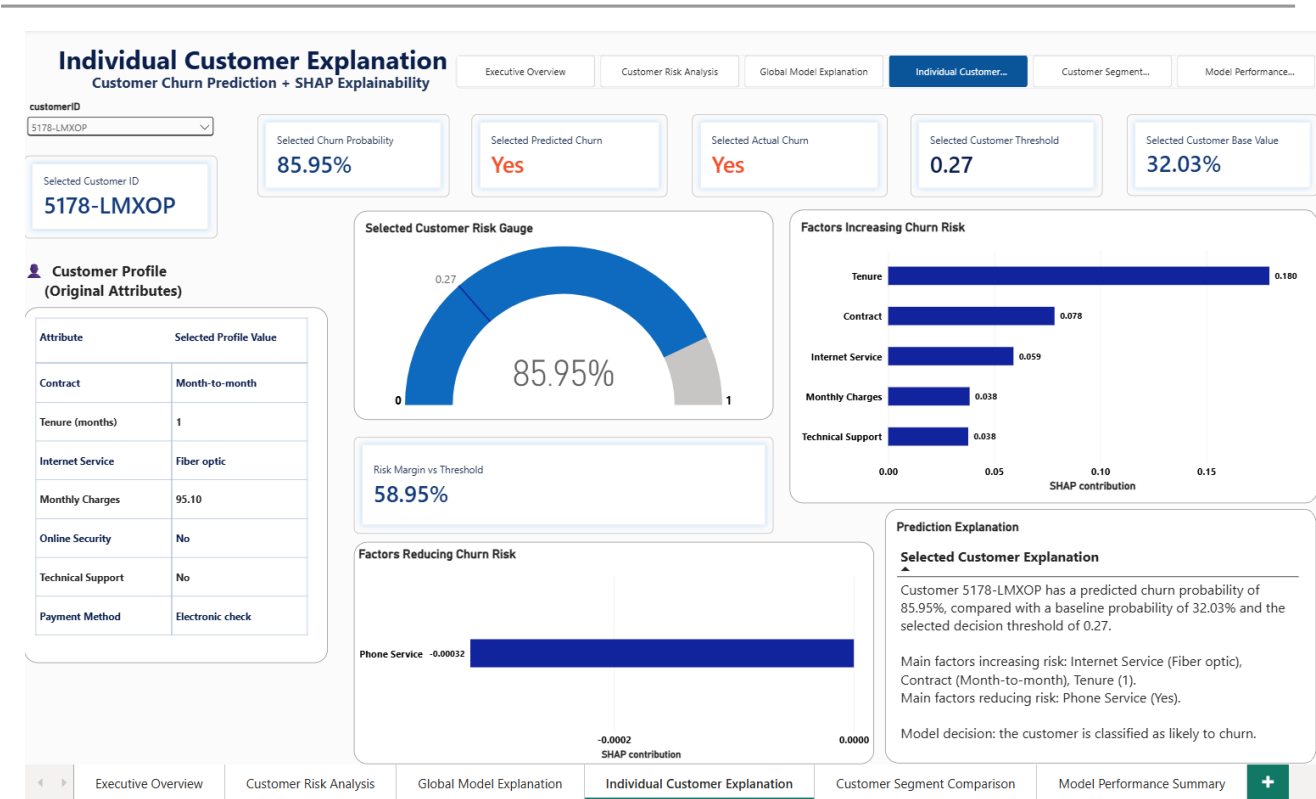


Figure 4. Fixed screenshot used in the static explanation condition.

Relevant task: Task 4 - Individual Customer Explanation

8. Static View 5 - Customer Segment Comparison

This view compares churn probability and churn rates across contract, internet service, payment method, senior-citizen status, and paperless billing.

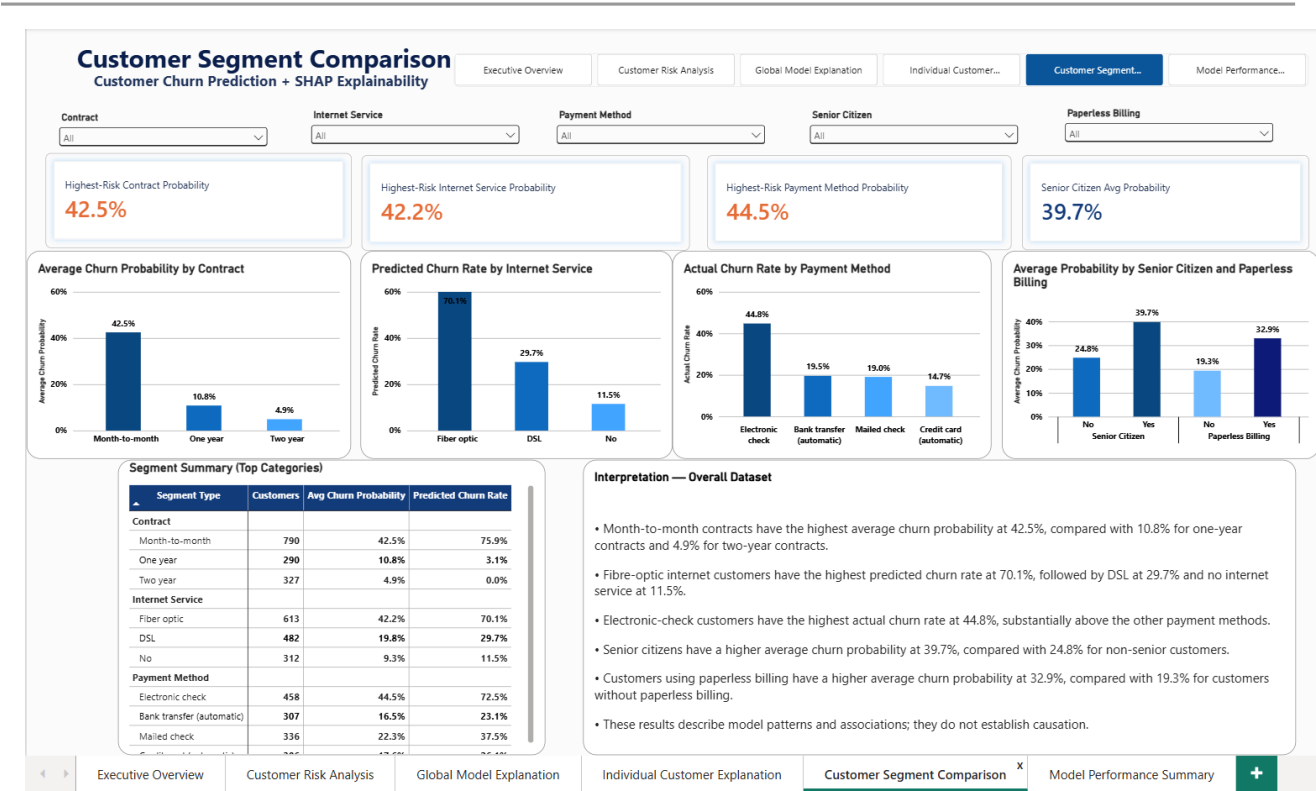


Figure 5. Fixed screenshot used in the static explanation condition.

Relevant task: Task 5 - Customer Segment Comparison

9. Static View 6 - Model Performance Summary

This view reports the final model metrics, compares the default and selected thresholds, and shows the reduction in false negatives.

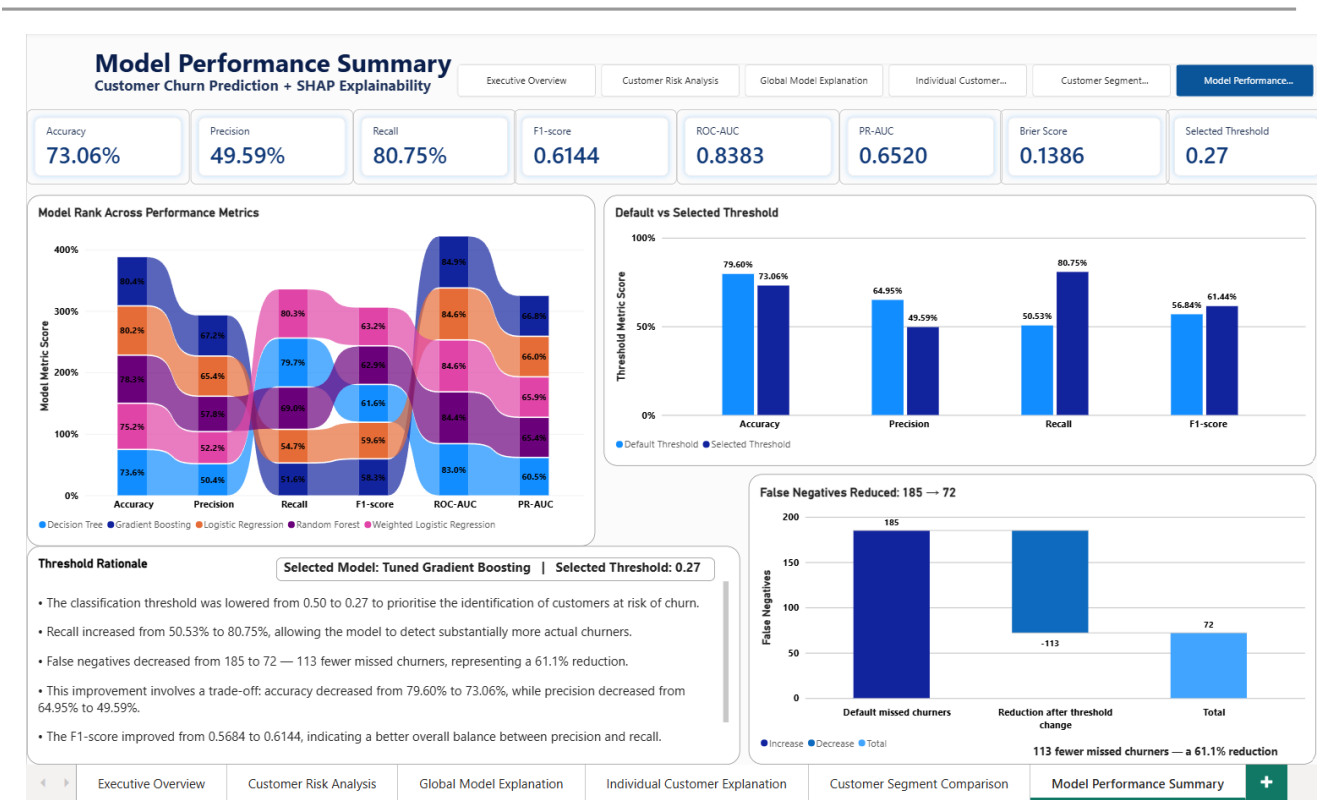


Figure 6. Fixed screenshot used in the static explanation condition.

Relevant task: Task 6 - Model Performance Summary

10. Condition A - Static Baseline Task Sheet

Use only the six fixed screenshots in this document. Record the total completion time after Task 6.

Task 1 - Executive Overview

- A. How many customers actually churned in the test dataset?
- B. How many customers were predicted to churn at the selected threshold?

Task 2 - Customer Risk Analysis

- A. Which customer has the highest churn probability?
- B. What is the displayed churn probability for that customer?

Task 3 - Global Model Explanation

- A. Which original customer feature has the highest global SHAP importance?
- B. Based on the Tenure Relationship chart, does longer tenure generally increase or reduce churn risk?

Task 4 - Individual Customer Explanation

- A. What is the predicted churn probability for customer 5178-LMXOP?
- B. Which factor has the strongest positive contribution to this customer's churn risk?

Task 5 - Customer Segment Comparison

- A. Which contract type has the highest average churn probability?
- B. Which internet-service category has the highest predicted churn rate?

Task 6 - Model Performance Summary

- A. What classification threshold was selected for the final model?
- B. By how many customers were false negatives reduced when the threshold changed from 0.50 to 0.27?

Static-condition completion time: _____ minutes / seconds

11. Condition B - Interactive Dashboard Task Sheet

Use the live Power BI report. Begin with all slicers reset to All and customer 5178-LMXOP selected on the individual explanation page. Record the total completion time after Task 6.

Task 1 - Executive Overview

- A. Navigate to Executive Overview. How many customers actually churned in the test dataset?
- B. How many customers were predicted to churn at the selected threshold?

Task 2 - Customer Risk Analysis

- A. Navigate to Customer Risk Analysis. Which customer has the highest churn probability?
- B. What is the displayed churn probability for that customer?

Task 3 - Global Model Explanation

- A. Navigate to Global Model Explanation. Which original customer feature has the highest global SHAP importance?
- B. Use the Tenure Relationship chart to determine whether longer tenure generally increases or reduces churn risk.

Task 4 - Individual Customer Explanation

- A. Select customer 5178-LMXOP. What is the predicted churn probability?
- B. Which factor has the strongest positive contribution to this customer's churn risk?

Task 5 - Customer Segment Comparison

- A. Navigate to Customer Segment Comparison. Which contract type has the highest average churn probability?
- B. Which internet-service category has the highest predicted churn rate?

Task 6 - Model Performance Summary

- A. Navigate to Model Performance Summary. What classification threshold was selected?
- B. By how many customers were false negatives reduced when the threshold changed from 0.50 to 0.27?

Interactive-condition completion time: _____ minutes / seconds

12. Answer Key and Scoring Guide

The same answer key is used for both conditions. Each correct Part A or Part B answer receives one point. The maximum task score is 12 points per condition.

Task	Correct Part A Answer	Correct Part B Answer
1	374 actual churn customers	609 predicted churn customers
2	5178-LMXOP	85.95% (86.0% accepted)
3	Tenure	Longer tenure generally reduces churn risk
4	85.95%	Tenure, with the largest positive SHAP contribution
5	Month-to-month contract (42.5%)	Fibre optic (70.1% predicted churn rate)
6	0.27	113 fewer false negatives, reduced from 185 to 72

13. Scoring Interpretation

- 10-12 points: Strong interpretation of the dashboard information
- 7-9 points: Generally correct interpretation with minor errors
- 4-6 points: Partial understanding of the presented information
- 0-3 points: Substantial difficulty locating or interpreting the information

Completion time should be analysed together with accuracy. A faster time is not treated as better when several answers are incorrect.

14. Phase 7.2 Summary

This document establishes the fixed baseline condition and the equivalent task set for the interactive dashboard. The next step is to prepare a short post-condition questionnaire and a structured response sheet for the ten participant profiles.